Normal Numbers

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Table of Contents

1. Basics

2. Normality and Uniform Distribution

3. Uncountable Sets

4. Examples

5. Rodney Nillsen's Work

6. Edmund Weitz's Work

Simple Normality

Definition

Let

$$\alpha = \sum_{j=-d}^{\infty} \alpha_{j,r} r^{-j}, \qquad \alpha_{j,r} \in \{0, 1, \ldots, r-1\}.$$

Define

$$p_{b,n,\alpha,r} = \#\{j \leq n : \alpha_{j,r} = b\}.$$

A number α is **simply normal** in base r if

$$\lim_{n \to \infty} \frac{p_{b,n,\alpha,r}}{n} = \frac{1}{r}$$

for every digit $b \in \{0, \ldots, r-1\}$.

Normal Numbers

- A real number α is normal in base $r$if every block of length k appears with frequency $\frac{1}{r^k}$.
- α is normal in base r if $r^m\alpha$ is simply normal in base r^n for all $m, n \geq 1$.

Definition
A number is **absolutely normal** if it is normal in every base $r > 1$.

What is Uniform Distribution Modulo 1?

Definition (Uniform Distribution Modulo 1)

A sequence of real numbers $(x_n)_{n\geq 1}$ is **uniformly distributed modulo 1** if for every subinterval $[u, v) \subseteq [0, 1)$,

$$\lim_{N\to\infty} \frac{\text{Card}\{1 \leq n \leq N : \{x_n\} \in [u, v)\}}{N} = v - u$$

where $\{x\} = x - \lfloor x \rfloor$ denotes the fractional part.

Theorem

Let $b \geq 2$ be an integer. The real number ξ is normal to base b if, and only if, the sequence $(\xi b^n)_{n\geq 1}$ is uniformly distributed modulo one.

Proof Setup: Connecting Digits to Fractional Parts

Goal: Assume ξ is normal to base b. We want to show (ξb^n) is uniformly distributed mod 1. Let $\xi = \lfloor \xi \rfloor + 0.a_1a_2a_3\ldots$ be the b-ary expansion of ξ.

- Let $D_k = d_1d_2\ldots d_k$ be a fixed block of k digits.
- Define the interval endpoints corresponding to this block:

$$u = \sum_{i=1}^{k} \frac{d_i}{b^i}, \quad v = \sum_{i=1}^{k} \frac{d_i}{b^i} + \frac{1}{b^k}$$

Key Observation

The block $a_na_{n+1}\ldots a_{n+k-1}$ equals $d_1d_2\ldots d_k$ **if and only if** $\{\xi b^{n-1}\} \in [u, v)$.

Proof: Counting and Limiting Behavior

Using our observation, the number of occurrences of the block D_k up to step N is exactly:

$$A_b(D_k, N, \xi) = \text{Card}\{1 \leq n \leq N - k + 1 : u \leq \{\xi b^{n-1}\} < v\}$$

Since ξ is normal to base b, by definition:

$$\lim_{N \to \infty} \frac{\text{Card}\{1 \leq n \leq N : u \leq \{\xi b^n\} < v\}}{N} = v - u = \frac{1}{b^k}$$

Proof: The Converse Direction ($\Leftarrow$)

Goal: Assume $(\xi b^n)_{n\geq 1}$ is uniformly distributed mod 1; show ξ is normal.

- Let u, v be arbitrary real numbers with $0 \leq u < v \leq 1$.
- Since general intervals aren't directly aligned with b-ary blocks, we approximate u and v from **inside and outside** using rational numbers u_1, u_2, v_1, v_2 whose denominators are powers of b:

$$u_1 \leq u \leq u_2 < v_2 \leq v \leq v_1 \quad \text{with } u_2 - u_1 < \varepsilon,\ v_1 - v_2 < \varepsilon$$

Sandwich Inequality

For any positive integer N, we bound our counting function:

$$\begin{aligned} \text{Card}\{1 \leq n \leq N, u_1 \leq \{\xi b^n\} < v_1\} \leq \\ \text{Card}\{\ldots, u \leq \{\xi b^n\} < v\} \leq \\ \text{Card}\{\ldots, u_2 \leq \{\xi b^n\} < v_2\} \end{aligned}$$

Proof: Conclusion via ε-Sandwiching

Dividing by N and taking the limit as $N \to \infty$:

$$v_1 - u_1 \leq \lim_{N\to\infty} \frac{\text{Card}\{1 \leq n \leq N, u \leq \{\xi b^n\} < v\}}{N} \leq v_2 - u_2$$

Using our bounds ($v_1 - u_1 < (v-u) + 2\varepsilon$ and $v_2 - u_2 > (v-u) - 2\varepsilon$):

$$v - u - 2\varepsilon \leq \lim_{N\to\infty} \frac{\text{Card}(\dots)}{N} \leq v - u + 2\varepsilon$$

- Since $\varepsilon > 0$ was arbitrary, the limit must equal $v - u$.
- This implies ξ is normal to base b. ■

Why Countability Fails on $[0, 1)$

Cantor's Diagonal Argument

The set of all infinite binary sequences,

$$2^{\mathbb{N}} = \{f : \mathbb{N} \longrightarrow \{0, 1\}\},$$

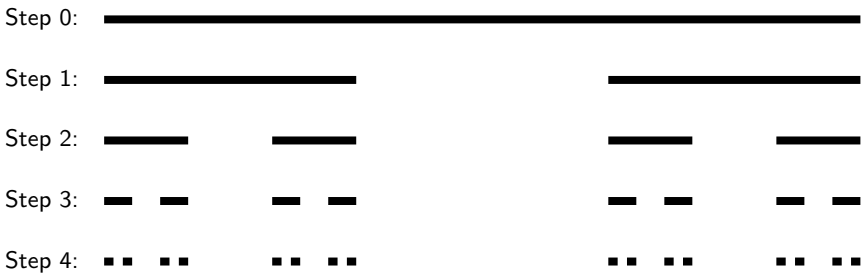
is **uncountable** (Shen & Vereshchagin, *Basic Set Theory*).

Why does this matter for analysis and probability?

- In discrete probability spaces (like coin flips over finite steps), we can assign probabilities to individual outcomes and simply add them up.
- On uncountable sets like $[0, 1)$, single points carry zero weight, so discrete summation breaks down—motivating **measure theory**.

Constructing the Cantor Set $\mathcal{C}$

Start with $[0, 1]$ and iteratively remove the open middle third of every remaining interval:



- **Ternary Characterization:** Removing middle thirds removes numbers whose base-3 expansion requires the digit 1.
- **Subtlety at Endpoints:** Endpoints like $\frac{1}{3}_{10} = 0.1\overline{0}_3 = 0.0\overline{2}_3$ remain in $\mathcal{C}$ because they admit a representation using **only 0's and 2's**.

Constructing an Onto Map $f : \mathcal{C} \longrightarrow [0,1)$

Every $x \in \mathcal{C}$ can be written in base 3 as $x = \sum_{k\in\mathbb{N}} \frac{a_k}{3^k}$ with $a_k \in \{0,2\}$.

The Cantor Surjection (Devil's Staircase Restriction)

Define $f : \mathcal{C} \longrightarrow [0,1)$ by replacing every 2 with a 1 and switching from base 3 to base 2:

$$f\left(\sum_{k\in\mathbb{N}} \frac{a_k}{3^k}\right) = \sum_{k\in\mathbb{N}} \frac{a_k/2}{2^k}$$

Concrete Example:

- Take $y = 0.\overline{1001}_2 \in [0,1)$ (which equals $\frac{9}{15} = \frac{3}{5}$ in base 10).
- Doubling each binary digit gives $x = 0.\overline{2002}_3 \in \mathcal{C}$ (which equals $\frac{56}{80} = \frac{7}{10}$ in base 10).
- Thus $f(x) = y$, showing every point in $[0,1)$ has a preimage in $\mathcal{C}$!

Uncountability of $[0, 1)$ & Need for Measure Theory

Logical Chain of Uncountability

1. $\mathcal{C}$ is in bijection with $2^{\mathbb{N}}$ (via $0 \leftrightarrow 0$ and $2 \leftrightarrow 1$), so $\mathcal{C}$ is **uncountable**.
2. Alternatively, $f : \mathcal{C} \twoheadrightarrow [0, 1)$ is surjective, and since $\mathcal{C} \subset [0, 1]$, the unit interval $[0, 1)$ is **uncountable**.

Measure-Theoretic Tools

Two fundamental facts:

Lemma 1

Every countable set has Lebesgue measure zero.

Lemma 2

A countable union of null sets is again a null set.

These facts allow us to show that the set of exceptional (non-normal) numbers is negligible from a measure-theoretic viewpoint.

What Does the Internet Think a Normal Number Is?

The Popular Conception

- A normal number is a real number whose digit expansion in any base is "uniform", meaning all finite digit strings occur with their statistically expected frequencies.
- If you randomly pick a number from Unif$(0, 1)$ (e.g., $0.5829017340192756\ldots$), it is normal with probability 1.

Rational Numbers

Rational Numbers ($x \in \mathbb{Q}$)

- Every rational number has a decimal (or b-ary) expansion that eventually **repeats periodically**.
- **Example 1:** $x = \frac{1}{5} = 0.2000\ldots$ (zeros completely dominate after the first digit).
- **Example 2:** $0.2437983983983983\ldots$ has a repeating tail.

Conclusion: Because their digits become periodic, rational numbers fail to contain all finite digit blocks with the required uniform asymptotic frequencies. Thus, **no rational number is normal**.

Liouville's Constant: Artificially Sparse Digits

Definition (Liouville's Constant)

$$\sum_{n=1}^{\infty} \frac{1}{10^{n!}} = 0.110001000000000000000001000\ldots$$

- **Why it fails normality:** The zeros overwhelmingly dominate the word length as n grows, because non-zero digits only appear at factorial positions ($n!$).
- **Modified Variant:** The same failure holds for its modified form $\sum_{n=1}^{\infty} \frac{n}{10^{n!}} = 0.120003000000\ldots$, which still suffers from massive gaps of zeros.

Are Famous Constants Normal?

Empirical Observations vs. Rigorous Proofs

- Analysis of the first trillion hexadecimal digits (base $b = 12$) of π shows digit 13 appears $62,49,96,13,666$ times, while others hover closely around $62,50,00,00,000$.
- Despite such compelling statistical evidence from massive computational searches, fundamental transcendental numbers like π and e **have not yet been rigorously proved to be normal**.

Normality is "easy" to find probabilistically (almost every number is normal), but exceedingly hard to verify deterministically for specific mathematical constants.

Binary Representation and Digit Counting

Just as base-10 expands decimals in powers of 10^{-i}, every $x \in [0,1)$ has a binary expansion (ending in zeros if dyadic):

$$x = \sum_{i=1}^{\infty} d_i 2^{-i} = \frac{d_1}{2^1} + \frac{d_2}{2^2} + \frac{d_3}{2^3} + \cdots, \qquad d_i \in \{0,1\}.$$

Let $S_n(x) = \sum_{i=1}^{n} d_i$ count the number of 1s among the first n binary digits of x.

Illustrative Examples

Number (x)	**Binary Expansion**	**Asymptotic Proportion $S_n(x)/n$**
$x = 0.5$	$0.1000\ldots_2$	$\to 0$
$x = 0.375$	$0.0110\ldots_2$	$\to 0$
$x = 1/3$	$0.010101\ldots_2$	$\to 1/2$

Simply Normal Numbers and Borel's Theorem

Definition (Simply Normal to Base 2)

A number $x \in [0,1)$ is *simply normal to base 2* if $\lim_{n\to\infty} \frac{S_n(x)}{n} = \frac{1}{2}$ (a 50/50 asymptotic balance of 0s and 1s).

Definition (Measure Zero & Almost Everywhere)

A set $A \subseteq [0,1)$ has *measure zero* if for every $\varepsilon > 0$ there exist intervals $J_1, J_2, \ldots$ covering A with $\sum_{n=1}^{\infty} \mu(J_n) < \varepsilon$ (e.g., all rationals in $[0,1)$, covered by intervals of length $\varepsilon/2^n$). A property holds *almost everywhere* if its complement has measure zero.

Theorem (Borel, 1904)

There exists a set $Z \subseteq [0,1)$ of measure zero such that every $x \in [0,1) \setminus Z$ is simply normal to base 2.

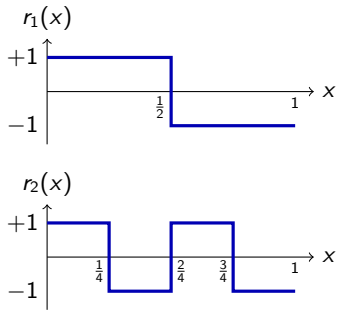
Rademacher Functions: Definition and Square Waves

Define $r_n(x) = +1$ if the n-th binary digit of x is 0, and -1 if it is 1. Equivalently, splitting $[0,1)$ into 2^n intervals of width $\frac{1}{2^n}$ ($k = 1, \ldots, 2^n$):

$$r_n(x) = \begin{cases} +1 & \text{if } x \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \text{ and } k \text{ is odd,} \\ -1 & \text{if } x \in \left[\frac{k-1}{2^n}, \frac{k}{2^n}\right) \text{ and } k \text{ is even.} \end{cases}$$

- $n = 1$ (2 parts): $k = 1$ (odd) $\Rightarrow +1$ on $[0, \frac{1}{2})$; $k = 2$ (even) $\Rightarrow -1$ on $[\frac{1}{2}, 1)$.
- $n = 2$ (4 parts): $+1$ on $[0, \frac{1}{4})$, $[\frac{2}{4}, \frac{3}{4})$ (odd k); -1 on $[\frac{1}{4}, \frac{2}{4})$, $[\frac{3}{4}, 1)$ (even k).

Square Wave Toggling on $[0,1)$



Rademacher Properties and Reformulation

Orthogonality Property (Integral of Distinct Products Is 0)

The integral of any product of distinct Rademacher functions over $[0,1)$ is exactly 0:

$$\int_0^1 r_{i_1}(x) r_{i_2}(x) \cdots r_{i_m}(x)\, dx = 0 \quad \text{for any distinct indices } i_1 < i_2 < \cdots < i_m.$$

Verification for $r_1(x)r_2(x)$ on $[0,1)$:

Interval	$r_1(x)$	$r_2(x)$	**Product** $r_1 r_2$
$[0, 0.25)$	$+1$	$+1$	$+1$
$[0.25, 0.50)$	$+1$	-1	-1
$[0.50, 0.75)$	-1	$+1$	-1
$[0.75, 1.00)$	-1	-1	$+1$

Area $= 0.25(+1 - 1 - 1 + 1) = 0$.

Reduction of Borel's Theorem

Setting $R_n(x) = \sum_{i=1}^{n} r_i(x) = n - 2S_n(x)$, $+1$s and -1s cancel when digits are balanced:

$$\lim_{n\to\infty} \frac{S_n(x)}{n} = \frac{1}{2} \iff \lim_{n\to\infty} \frac{R_n(x)}{n} = 0.$$

Step 1: Fourth-Power Moment

Since $\int_0^1 r_{i_1} \cdots r_{i_m}\, dx = 0$ for distinct indices and $r_i^2 = 1$, expanding $R_n^4 = \sum_{i,j,k,l=1}^{n} r_i r_j r_k r_l$ leaves only two non-vanishing index patterns:

Case	Pattern	Number of Terms	Integral
C-1	r_i^4	n	1
C-2	$r_i^2 r_j^2$ $(i \neq j)$	$3n(n-1)$	1
C-3–C-5	$r_i^3 r_j,\ r_i^2 r_j r_k,\ r_i r_j r_k r_l$	$n^4 - 3n^2 + 2n$	0

Consequently, $\int_0^1 \left(\frac{R_n(x)}{n}\right)^4 dx = \frac{3n^2-2n}{n^4} = \frac{3n-2}{n^3} = \frac{3}{n^2} - \frac{2}{n^3}$. By the p-series test $(p = 2, 3 > 1)$:

$$\sum_{n=1}^{\infty} \int_0^1 \left(\frac{r_1(x) + \cdots + r_n(x)}{n}\right)^4 dx = \sum_{n=1}^{\infty} \left(\frac{3}{n^2} - \frac{2}{n^3}\right) < \infty.$$

Beppo Levi's Theorem and Finite Area Intuition

Beppo Levi's Theorem (Interchanging Sum and Integral)

Because each fourth-power term $\left(\frac{R_n(x)}{n}\right)^4 \geq 0$ is always non-negative, we are allowed to interchange summation and integration:

$$\int_0^1 \left[\sum_{n=1}^{\infty} \left(\frac{r_1(x) + \cdots + r_n(x)}{n}\right)^4\right] dx = \sum_{n=1}^{\infty} \int_0^1 \left(\frac{r_1(x) + \cdots + r_n(x)}{n}\right)^4 dx < \infty.$$

Why Finite Area Implies a Finite Function Almost Everywhere

- **Finite Total Area:** The total area under the curve of this infinite sum function over $[0, 1)$ is finite.
- **No Blow-Up on Any Interval:** If the total area under a curve is finite, the function itself cannot blow up to infinity over any real interval.
- **Almost-Everywhere Convergence:** Hence the infinite sum converges to a finite number for almost all $x \in [0, 1)$.

Step 2 & Step 3: Multiplier Lemma and Step-Function Convergence

Lemma (Step 2: Multiplier Lemma)

If $a_n \geq 0$ and $\sum_{n=1}^{\infty} a_n < \infty$, there exist $b_n > 0$ with $\lim_{n\to\infty} b_n = \infty$ and $\sum_{n=1}^{\infty} a_n b_n < \infty$.

Theorem (Step 3: Almost-Everywhere Convergence of Step Functions)

Let (φ_n) be step functions on $[0,1)$ with $\sum_{n=1}^{\infty} \int_0^1 |\varphi_n(x)|\, dx < \infty$. Then

$$\lim_{n\to\infty} \varphi_n(x) = 0 \quad \text{for almost all } x \in [0,1).$$

Intuition

When the total integrated area $\sum_{n=1}^{\infty} \int_0^1 |\varphi_n(x)|\, dx$ is finite, the functions $\varphi_n(x)$ must shrink to 0 at almost every point $x \in [0,1)$.

Step 3: Proof Sketch

1. **Apply Multiplier Lemma:** Set $a_n = \int_0^1 |\varphi_n(x)|\,dx$. Since $\sum a_n < \infty$, choose $b_n \to \infty$ such that $\sum_{n=1}^{\infty} b_n a_n < \infty$.

2. **Bad Set & Threshold:** Let $Z = \{x \in [0,1) : \varphi_n(x) \not\to 0\}$. For $x \in Z$, there exists $\eta > 0$ with $|\varphi_n(x)| > \eta$ infinitely often. Since $b_n^{-1} \to 0$, we have $b_n^{-1} < \eta$ for large n, so x enters

$$A_n = \left\{x \in [0,1) : |\varphi_n(x)| > b_n^{-1}\right\}$$

for infinitely many n (each A_n is a finite union of flat intervals).

3. **Forward Union Trap:** Every $x \in Z$ belongs to the forward union starting from any index $N \geq 1$:

$$Z \subseteq \bigcup_{k=N}^{\infty} A_k.$$

4. **Measure Bound & Tail Limit:** On A_n, $b_n|\varphi_n(x)| > 1$, so

$$\mu(A_n) = \int_{A_n} 1\,dx \leq \int_0^1 b_n|\varphi_n(x)|\,dx = b_n a_n.$$

Thus $\mu(Z) \leq \sum_{k=N}^{\infty} \mu(A_k) \leq \sum_{k=N}^{\infty} b_k a_k \to 0$ as $N \to \infty$, proving $\mu(Z) = 0$. □

Step 4: Proof of Borel's Theorem

Define the non-negative step functions on $[0,1)$:

$$\varphi_n(x) = \left(\frac{R_n(x)}{n}\right)^4 = \left(\frac{r_1(x) + \cdots + r_n(x)}{n}\right)^4 \geq 0.$$

- **Finite Total Area (Step 1):** Each φ_n is a step function on $[0,1)$ and

$$\sum_{n=1}^{\infty} \int_0^1 |\varphi_n(x)|\, dx = \sum_{n=1}^{\infty} \left(\frac{3}{n^2} - \frac{2}{n^3}\right) < \infty.$$

- **Almost-Everywhere Convergence (Step 3):** Applying the Step-Function Convergence Theorem gives $\lim_{n\to\infty} \varphi_n(x) = 0$ for almost all $x \in [0,1)$.

- **Fourth Root & Digit Ratio:** Taking fourth roots and substituting $R_n(x) = n - 2S_n(x)$ yields

$$\lim_{n\to\infty} \frac{R_n(x)}{n} = 0 \implies \lim_{n\to\infty} \frac{S_n(x)}{n} = \frac{1}{2} \quad \text{for almost all } x \in [0,1). \quad \square$$

Concluding Remark: Full Normality to Base 2

Definition (Full Normality to Base 2)

A number $x \in (0,1)$ is *normal to base 2* (stronger than simply normal) if every finite binary block $b_1 b_2 \dots b_r$ of length r occurs with asymptotic frequency 2^{-r}:

$$\lim_{n\to\infty} \frac{1}{n} |\{1 \le j \le n : d_j(x) = b_1, \dots, d_{j+r-1}(x) = b_r\}| = \frac{1}{2^r}.$$

Why Full Normality Is Stronger ($r = 2$, $x = 1/3 = 0.010101\ldots_2$):

- For $r = 2$, there are $2^2 = 4$ possible binary blocks (00, 01, 10, 11), each required to appear with frequency $\frac{1}{4}$ (25%).
- For $x = 1/3$, single digits have a 50/50 balance (*simply normal*), yet 01 and 10 appear 50% of the time while 00 and 11 never appear (0%) $\implies$ **not fully normal**.

Hausdorff's Contribution

The Historic Result

In 1914, Felix Hausdorff published an elegant, beautifully constructive proof showing that **almost all numbers are simply normal in base 2**.

Lemma

If r is an integer greater than 1, there is a positive constant D such that for all $n \in \mathbb{N}^+$:

$$\sum_{p=0}^{n} \binom{n}{p} \frac{(r-1)^{n-p}}{r^n} \left(\frac{p}{n} - \frac{1}{r}\right)^4 \leq \frac{D}{n^2}$$

Proof Strategy: The Differential Operator

We fix positive integers s and n, and define functions recursively:

$$f_0(x,y) = \sum_{p=0}^{n} \binom{n}{p} x^{sp} y^{n-p} = (x^s + y)^n$$

$$f_{k+1}(x,y) = x \cdot \frac{\partial}{\partial x} f_k(x,y) - y \cdot \frac{\partial}{\partial y} f_k(x,y)$$

By working with individual summands, one can show:

$$f_k(x,y) = \sum_{p=0}^{n} \binom{n}{p} \big((s+1)p - n\big)^k x^{sp} y^{n-p}$$

Goal: Compute $f_4(x,y)$ explicitly from $(x^s + y)^n$ to evaluate the fourth moment.

Evaluating f_4 and Setting Parameters

Expanding $f_4(x, y)$ yields a polynomial of the form:

$$f_4(x, y) = n(x^s + y)^{n-4} Q$$

where Q groups terms containing powers of n and s.

The Clever Substitution

Set $r = s + 1$, $x = 1/\sqrt[s]{r}$, and $y = (r - 1)/r$.

- This implies $sx^s - y = 0$, causing several terms in Q to vanish.
- The remaining terms simplify to a clean second-degree polynomial in n:

$$f_4(x, y) = 3(r - 1)^2 n^2 + (r^3 - 7r^2 + 12r - 6)n \leq Cn^2$$

From Lemma to Measure of Deviations

The measure of intervals starting with a specific sequence of length n containing p occurrences of digit b is:

$$\lambda = \binom{n}{p}(r-1)^{n-p}\frac{1}{r^n}$$

Define $M_b(n,\epsilon)$ as the set of numbers where relative frequency deviates by $\geq \epsilon$:

$$M_b(n,\epsilon) = \left\{\alpha \in [0,1) : \left|\frac{p}{n} - \frac{1}{r}\right| \geq \epsilon\right\}$$

Using our lemma:

$$\epsilon^4\lambda(M_b(n,\epsilon)) \leq \sum \binom{n}{p} \ldots \left(\frac{p}{n} - \frac{1}{r}\right)^4 \leq \frac{D}{n^2} \implies \lambda(M_b(n,\epsilon)) \leq \frac{D}{\epsilon^4 n^2}$$

Applying Borel-Cantelli Lemma

Let $M_b(\epsilon)$ be the set where frequency deviates by $\geq \epsilon$ for **infinitely many** n:

$$M_b(\epsilon) = \bigcap_{m=1}^{\infty} \bigcup_{n=m}^{\infty} M_b(n, \epsilon) = \bigcap_{m=1}^{\infty} S_b(m, \epsilon)$$

- Summing the tail bounds:

$$\lambda(S_b(m, \epsilon)) \leq \sum_{n=m}^{\infty} \lambda(M_b(n, \epsilon)) \leq \frac{D}{\epsilon^4} \sum_{n=m}^{\infty} \frac{1}{n^2}$$

- Since $\sum 1/n^2$ converges, choosing m large makes the measure arbitrarily small.
- Thus, $M_b(\epsilon)$ is a **null set**.

From Simply Normal to Absolutely Normal

Theorem

If $r \geq 2$ is an arbitrary base, then **almost all numbers are simply normal** *in this base.*

- The set of exceptional numbers for a single digit b is $M_b = \bigcup_{k=1}^{\infty} M_b(1/k)$ (countable union of null sets $\implies$ null set).
- Across all r digits, the set of non-simply normal numbers is a finite union of null sets.

Corollary (Normal in Base r)

Almost all numbers are normal in base $r \geq 2$.

Corollary (Absolute Normality)

Since there are only countably many bases r, a countable union of null sets implies **almost all numbers are absolutely normal**.

Questions?